MIL-G-17489B(SHIPS)
15 March 1961
SUPERSEDING
MIL-G-17489A(SHIPS)
13 August 1959

MILITARY SPECIFICATION

GAGES, PRESSURE (AIR), BOILER

1. SCOPE

1.1 Scope. - This specification covers boiler air pressure gages suitable for measuring air pressure up to and including 200 inches of water.

1.2 Classification. - Air pressure gages shall be of the following types and classes as specified (see 6.2):

Type A - Dial.
Type B - Vertical scale.
  Class 1 - Single unit.
  Class 2 - Dual unit.
  Class 3 - Triple unit.
  Class 4 - Multiple unit.
Type C - Window.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for.
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment, Naval Shipboard Use.
MIL-P-15024 - Plates, Identification Information and Marking for Identification of Electrical, Electronics and Mechanical Equipment.
MIL-E-15090 - Enamel Equipment, Light-Gray, (Formula No. 111).
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction. - Prior to production, one air pressure gage of each type and class (except type C), shall be subjected to the tests specified in 4.5 and 4.6 subject to the requirements of 4.2 and 4.2.1. Satisfactory completion of testing of a class 2 gage under type B will indicate ability to furnish all classes of type B gages.

3.2 Material. - Material shall be of the best commercial quality and as specified herein for each type and class of pressure gage.

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3.3 Movement. - All parts of the pressure gage movement shall be of corrosion-resisting materials, or shall be protected by suitably treating or coating the surfaces. All moving parts shall be as light as practicable, consistent with 3.3.1 and 3.3.2.

3.3.1 All wearing surfaces shall be of metals which insure minimum friction and wear.

3.3.2 The parts subjected to wear shall be of alloy steel, heat treated or surface hardened, or of a nonferrous metal.

3.4 Mechanism. - The mechanism shall be designed and constructed so that no leakage will occur when the pressure gage is subjected to maximum working pressure at a temperature not in excess of 145°F.

3.5 Stopcock. - Each pressure gage shall be provided with a three-way stopcock which will enable checking the zero setting at any time without breaking connection. The stopcock shall be located as near as possible to the threaded connection of the pressure gage. Each pressure gage shall be provided with a flared union coupling suitable for attaching to the stopcock.

3.6 Connection. - Pressure gages shall have a 1/4 inch male pipe thread connection, threaded in conformance with Handbook H28. The connection shall be at the bottom, unless otherwise specified (see 6.2). The connection shall extend beyond the case not less than 1-1/8 inches in bottom connected pressure gages and not less than 1-1/4 inches beyond the mounting bosses of back connected pressure gages. The minimum thread length shall be 1/2 inch.

3.7 Repeatability. - Pressure gages shall be of such quality as to insure repeatability within one scale division for the entire scale range.

3.8 Shock and vibration. - All types and classes (except type C) of pressure gages shall satisfactorily withstand HI shock in accordance with Specification MIL-S-901 and vibration in accordance with Standard MIL-STD-167.

3.9 Design. - Pressure gages shall be mechanically actuated through the use of diaphragms or other devices without the use of any liquids. The movement of the diaphragms due to differences in pressure shall be magnified by means of a pointer connected to the diaphragms by links of gearing. Design of pressure gage and materials selected shall be adequate to insure a maximum pressure gage service life of ten years, minimum pressure gage life of five years.

3.10 Pointers. - Pressure gages shall be fitted with indicating pointers of a plain and unembellished design (see figure 1). Micrometer adjustable design similar to that shown on figure 1, or other zero adjustment feature of the indicating pointer is required so that the scale may be adjusted 5 degrees or 1/2 inch, whichever is greater, in both directions from the normal position. Finished pointers shall be in near mechanical balance. Black pointers shall be furnished. Pressure gages shall be provided with stops at both ends of the diaphragm drive to prevent jamming of the pointer.

3.11 Dials and scales. - The dials and scales for the pressure gages shall be white in appearance with black graduation, figures and letters. Marking shall be neatly and accurately lithographed, silk-screened, cut or etched and filled with dead black enamel. Dials and scales for gages with a pointer shall be made of sheet steel or aluminum adequately protected from corrosion. Scale markings shall be of sufficient size to be seen by normal eyes at a distance of 20 feet except for those intended for control panel installation (see 3.15.1) and except as modified in 3.16.4 for type C gages.

3.12 Cases. - Cases shall be made of sheet steel or aluminum adequately protected from corrosion. Cases shall be free from warpage, checks and chipped or blistered surfaces. There shall be no burrs or projecting ridges at the dividing line of mold, nor an unsightly finish at this point caused by chipping, filing, or grind without subsequent buffing or polishing. Cases shall be painted gray conforming to class 2 of Specification MIL-E-15090, unless otherwise specified in the contract or order.

3.13 Identification plates. - Identification plates shall be furnished in accordance with Specification MIL-P-15024 with each gage either mounted on the gage or for mounting on the panel board directly under the pressure gage. The identification plate shall show the purpose for which the pressure gage is intended (see 6.2).

3.13.1 In addition to the identification plates specified in 3.13 each pressure gage shall be marked with the manufacturer's name or trademark.

3.14 Type A, dial. -

3.14.1 Dials. - Dials shall be as shown on figure 2 in respect to character and design. Pressure gages shall be accurately and correctly graduated up to and including the last scale division of the scale. The graduation shall cover an arc not less than 270 degrees. Dials shall be secured to the pressure gage mechanism by at least 3 screws. The scales shall be graduated to read the pressure effective on the diaphragm in inches of water above atmosphere (barometric) unless other graduations are specified.

3.14.1.1 The diameter of the dial, height of the figures and diameter of graduated dial circle thereon, shall be as shown in table I.

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Table I - Dimensions of type A dials.

Size of pressure gage	Diameter of dial	Height of figures	Diameter of graduated dial circles minimum[1]
Inches	Inches	Inch	Inches
8-1/2	8-1/4	1/2	7

[1]The diameter of the dial circle shall be slightly smaller than the opening in the bezel ring.

3.14.1.2 Unless otherwise specified in the contract or order, the graduations on the dials shall be as shown in table II.

Table II - Graduation of type A dials.

Range	Linear scale 0-90 degrees	Linear scale 90-270 degrees
Inches	Inches	Inches
0-140	0-10	10-100
0-170	0-15	15-150
0-200	0-15	15-200

3.14.1.3 The first 90 angular degrees of the dial arc shall cover the inches of water specified in table II divided into 2/10 inches of water with numerals every inch. The 1 inch and 2/10 inch scale graduations shall be easily distinguished from one another. The remainder of the scale shall cover at least 180 angular degrees of the dial arc with sufficient numerals and scale graduations for good and accurate readibility.

3.14.2 Actuating mechanism. - The actuating mechanism between the pointer and the diaphragms shall be such that the pointer will move in an arc of a circle not less than 270 degrees.

3.14.3 Accuracy. - Type A pressure gages shall be of such design and construction as to insure maintenance of accuracy to within plus or minus one scale division for the 90 degree portion of the scale range, and for the remainder of the scale range, within plus or minus 2 percent of the total scale range (see 4.5.3).

3.14.4 Cases. - Cases shall conform to the design requirements specified in 3.12. In addition, design character and dimensions shall be as shown on figure 3. Dimensions other than those for the mounting bolt holes shall be considered nominal, and reasonable variations will be permitted.

3.14.5 Movement. - Movement shall be secured to the socket in a manner independent of the case.

3.14.6 Sockets. - Sockets shall be made of bronze or brass. Sockets shall be provided with an integral wrench grip of 1/2 inch minimum width outside the case, threaded for the iron pipe size connection. The socket shall be secured to the case by not less than 3 screws in such a manner that working or twisting of the pressure gage case shall not cause movement of the pointer.

3.15 Type B, vertical scale. -

3.15.1 Pressure gage scales. - Scales shall be as shown on figure 4 in respect to character and design except for ranges and number of units. Pressure gages shall be accurately and correctly graduated up to and including the last division of the scale. Scales shall be secured to the pressure gage mechanisms by at least 4 screws. The scales shall be of material as specified in 3.11. The scales shall be graduated to read the pressure effective on the diaphragm in inches of water above atmosphere (barometeric) unless other graduations are specified in the contract or order. The height of figures on the scale shall be 1/2 inch. Scales with a range of up to and including 15 inches of water shall have numerals at each inch of water. Each inch graduation shall be divided into 1/10 inch graduation. The 1 inch, 1/2 inch and 1/10 inch graduations shall be easily distinguished from each other. Scales with a range up to and including 60 inches shall have numerals every 5 inch graduation. Scales up to and including 100 inches shall have numerals every 10 inch graduation. The graduation shall be in inches with the 10 inch, 5 inch and 1 inch graduations easily distinguished from each other. Scales with a range above 100 inches of water shall be marked with sufficient graduations for good and accurate readability. Minimum size of scale shall be 2 inches by 10 inches, except that for gages intended for control panel installation, scales may be a minimum of 2 inches by 6 inches.

3.15.2 Actuating mechanism. - The actuating mechanism between the pointer and the diaphragms shall be such that the pointer shall move in a vertical plane for a minimum distance of 10 inches, except that for control panel installation this minimum distance may be 6 inches.

3.15.3 Accuracy. - Gages shall be of such design and construction to insure maintenance of accuracy to within plus or minus 2 percent of total scale range (see 4.5.3).

3.15.4 Type B, class 1, vertical scale, single unit. - The gage shall consist of a single case containing one indicating scale. The range of scale

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shall be as specified (see 6.2), and shall be the next increment divisible by 5 above the total draft loss through the boiler at overload condition.

3.15.5 Type B, class 2 vertical scale, dual unit. - The gage shall consist of a single case containing two indicating scales, one low range and the other high range. The ranges of the scales shall be as specified (see 6.2). There shall be no overlapping of the range scales except that the high range scale shall have a zero graduation for zero adjustment of scale and then blanked-off up to the top graduation of the low range scale. The low range scale is to cover total draft losses through the boiler at cruising power and shall be the next increment divisible by 5 above this total. The high range scale is to cover total draft losses through the boiler at full power and overload conditions, the specified range being the next increment divisible by 5 above the total draft loss at boiler overload condition.

3.15.6 Type B, class 3, vertical scale triple unit. - The gage shall consist of a single case containing three indicating scales, one low, one intermediate and one high range. The ranges shall be as specified (see 6.2). There shall be no overlapping of the range scales except that the intermediate and high range scales shall have a zero graduation for zero adjustment of the scales. The intermediate range scale shall be blanked-off to the top graduation of the low range scale and the high range scale shall be blanked-off to the top graduation of the intermediate scale. The low range scale is to cover total draft losses through the boiler at cruising power and shall be selected as specified in 3.15.5. The intermediate range scale is to cover total draft losses through the boiler at full power conditions, the specified range being the next increment divisible by 5 above the total draft loss at boiler full power condition. The high range scale is to cover total draft losses through the boiler at overload power conditions, the specified range being selected as indicated in 3.15.5.

3.15.7 Type B, class 4, vertical scale multiple unit. - The gage shall consist of a single case containing not more than 4 scales. This case may contain four class 1 units or 2 class 2 units. The ranges shall be as specified (see 6.2).

3.15.8 Cases. - Gage installations may employ any practicable arrangement of cases within the maximum panel openings specified in table III. Each case shall be fitted with clear glass or clear plastic, free from blemishes and held securely in place. Unless otherwise specified (see 6.2), cases shall conform to figure 5 and have a maximum depth and front extension as shown thereon, and shall be suitable for mounting in panel openings of 13-5/8 inches maximum height and maximum width as shown in table III. A maximum of 4 scales shall be contained in a case.

Table III - Maximum width of panel opening.

Number of indicating scales	Maximum width of panel opening
	Inches
1	4
2	6-3/4
3	9-1/4
4	12

3.16 Type C window. -

3.16.1 Unless otherwise specified (see 6.2), the gage shall be as shown on figure 6. The actuating mechanism between the pointer and the diaphragms shall be such that the pointer shall move in an arc of a circle not less than 65 degrees and not over 180 degrees. The scale, pointer and actuating mechanism shall be completely enclosed in a case with a heat-treated glass or a thermosetting plastic front, to permit observation of the pointer and scale. Dials shall be as shown on figure 6. Gages shall be accurately and correctly graduated up to and including the last division of the scale. The graduation shall cover an arc of not less than 65 degrees. The dial shall be secured to the case by screws. Unless otherwise specified (see 6.2), each type C pressure gage shall be furnished with the following 4 sizes of flared union couplings:

Inches, O.D.
0.250
.405
.500
.540

3.16.2 Gage scales. - Unless otherwise specified (see 6.2), the range of the scale shall be one of the following:

(a) 0 inches to 20 inches inclusive.
(b) 0 inches to 50 inches inclusive.

3.16.3 Accuracy. - The gages shall be of such design and construction as will assure maintenance of accuracy to within plus or minus 2 percent of total scale range (see 4.5.3).

3.16.4 Cases. - Cases shall conform to the requirements of 3.12 except that the front can be die-cast zinc adequately protected from corrosion. In figure 6, dimensions other than those for the mounting holes shall be considered nominal and reasonable variations are permitted. The maximum overall dimensions of the gage in any drawing shall not exceed 20 inches.

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3.17 Drawings. - Drawings as specified (see 6.2), shall be furnished in accordance with Specification MIL-D-963.

3.18 Repair parts. -

3.18.1 Repair parts shall be furnished in quantities as specified (see 6.2). Each pressure gage shall be furnished with two diaphragm sets which shall be packaged to include detail instructions as to method of diaphragm replacement. Repair parts shall be as approved for each supplier by the bureau or agency concerned and shall consist of sufficient parts to maintain the instrument for a minimum of one year.

3.18.2 Onboard. - Onboard repair parts shall be marked in accordance with Standard MIL-STD-129 and to show the identification numbers corresponding to those shown on the repair parts list and the standard Federal stock number for that part.

3.18.3 Stock. - Stock repair parts shall be furnished in accordance with Specification MIL-P-15137.

3.19 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction (except type C). - One air pressure gage of each type and class (except type C) shall be forwarded to the U.S. Naval Boiler and Turbine Laboratory, Philadelphia, Pa. or to any other laboratory satisfactory to the Bureau of Ships for the tests specified in 4.5. In addition, a unit or units shall be service tested by shipboard installation as specified in 4.6. After approval of the basic unit, the tests need not be repeated unless there are changes in the design or construction of the pressure gage (see 3.1).

4.2.1 Where a manufacturer has completed the tests specified in 4.2 on pressure gages required by this specification and the pressure gages have been accepted by the Bureau of Ships, a test report of the tests conducted will be acceptable as proof of satisfactory performance.

4.3 Sampling. -

4.3.1 Lot. - All pressure gages of the same type and class offered for delivery at one time under the same contract or order shall be considered a lot for purposes of sampling and inspection.

4.3.2 Sampling for visual and dimensional examination. - A random sample of pressure gages shall be selected from each lot in accordance with Standard MIL-STD-105 at inspection level II for the examination specified in 4.4.2. The acceptance quality level shall be 1.5 percent defective.

4.3.3 Sampling for lot acceptance tests. - A random sample of pressure gages shall be selected from each lot in accordance with Standard MIL-STD-105 at inspection level L-6 for the tests specified in 4.5.1 through 4.5.5. For each test the acceptable quality level shall be 1.5 defective; however, the sample sizes shall be as specified in the applicable table of Standard MIL-STD-105 and the acceptance number shall be zero until the AQL permits one or greater.

4.4 Examination. -

4.4.1 Examination and calibration. - Before submission for the examination specified in 4.4.2, and in accordance with 4.5.3, the contractor shall examine and calibrate each gage.

4.4.2 Visual and dimensional examination. - Each of the sample gages selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification as to material, finishes, workmanship, dimensions and scale or dial markings. Any gage in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective gages in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Test procedures. -

4.5.1 Temperature test. - The pressure gages shall be subjected to a temperature of 145°F. for a period of not less than 6-hours.

4.5.2 Excess pressure test. - Upon completion of the temperature test, a pressure equal to 1-3/10 times the maximum scale reading shall be applied for 1 hour. Within 10 minutes after the release of this pressure, and without any adjustment, the gage shall be subjected to the calibration test specified in 4.5.3.

4.5.3 Calibration test. - Calibration tests reading both up and down shall be made to determine conformance with 3.14.3, 3.15.3, or 3.16.3, as applicable.

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4.5.3.1 The pressure gage shall be tested for accuracy of indication at approximately two-thirds of full scale and at zero pressure.

4.5.3.2 The gage shall be tested for accuracy of indications at all major scale markings with increasing pressure and also with decreasing pressure.

4.5.4 Leakage test. - Leakage test shall be conducted to determine conformance with 3.4.

4.5.5 Roll and pitch test. - The gages shall be subjected to a roll and pitch test. Simulation of roll and pitch of a ship shall be made by suspending the gage from a pendulum mounting and subjecting it to a swing of 45 degrees each side of the vertical with a period of roll of one complete cycle in 12 seconds.

4.5.6 Shock test. - Shock tests shall be conducted (except for type C) in accordance with type A, grade I, equipment of Specification MIL-S-901 (see 3.8).

4.5.7 Vibration test. - Vibration tests (except for type C) shall be conducted in accordance with Standard MIL-STD-167 (see 3.8).

4.6 Service test. - Service tests shall be made by a shipboard installation to determine conformance with 3.9 for a period of not less than 6 months or a period determined by the Bureau of Ships.

4.7 Sample items, packages, and packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance to Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment--material for early equipment installation and for short term storage of onboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be unit protected and packaged in accordance with Specification MIL-P-116. Repair parts shall be packaged one part to a unit package unless used in sets. Unit containers, except for those required by Specification MIL-P-116 for the applicable method of preservation, shall conform to the following boxes at the option of the supplier:

Specification	Box
PPP-B-566	Folding, paperboard
PPP-B-636	Fiberboard
PPP-B-665	Paperboard, metal stayed
PPP-B-676	Set-up paperboard

Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.1.2.1 Packing. - Onboard repair parts shall be packed in accordance with 5.1.1.2.

5.1.2.2 Marking. - Interior packages and exterior shipping containers shall be marked for shipment in accordance with Standard MIL-STD-129.

5.2 Domestic shipment and long term storage or overseas shipment requirement. - The requirement and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment for long term storage or overseas shipment for use when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. -

5.2.1.1.1.1 Gages shall be individually unit protected and packaged in accordance with method IA of Specification MIL-P-116. Unless otherwise specified, the selection of the sub-method of unit protection shall be at the option of the contractor. Except for those unit containers required under Specification MIL-P-116 for the applicable method of preservation, each gage shall be unit packaged in a fiberboard box conforming to Specification PPP-B-636. All openings of the gage shall be sealed with pressure-sensitive tape conforming to Specification PPP-T-60.

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5.2.1.1.1.2 Repair parts. - Repair parts shall be unit protected and packaged in accordance with Specification MIL-P-116. Unless otherwise specified, parts shall be packaged one part to a unit package unless used in sets. Unit containers, except for those required by Specification MIL-P-116 for the applicable method of preservation, shall conform to the following boxes at the option of the supplier:

Specification	Box
PPP-B-566	Folding, paperboard
PPP-B-636	Fiberboard
PPP-B-665	Paperboard, metal stayed
PPP-B-676	Set-up paperboard

Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.2.1.1.2 Level C. - Preservation and packaging of gages shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.3 Packing. -

5.2.1.3.1 Basic equipment. -

5.2.1.3.1.1 Level A. - Gages, packaged as specified (see 5.1), shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Box	Type or class
PPP-B-585	Wood, wirebound	Class 3 use
PPP-B-591	Fiberboard, wood-cleated	Overseas type
PPP-B-601	Wood, cleated-plywood	Overseas type
PPP-B-621	Wood, nailed and lock corner	Class 2
PPP-B-636	Fiberboard	Class 2
MIL-B-10377	Wood-cleated, veneer paper overlaid	Overseas type

Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Caseliners for class 2 fiberboard boxes conforming to Specification PPP-B-636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 2 inch wide tape conforming to Specification PPP-T-76 or PPP-T-60. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.1.3.1.2 Level B. - Gages, packaged as specified (see 5.1), shall be packed in boxes conforming to any of the following specifications at the option of the supplier:

Specification	Box	Type or class
PPP-B-585	Wood, wirebound	Class 1 or 2 use
PPP-B-591	Fiberboard, wood-cleated	Domestic type
PPP-B-601	Wood, cleated-plywood	Domestic type
PPP-B-621	Wood, nailed and lock corner	Class 1
PPP-B-636	Fiberboard	Class 2
MIL-B-10377	Wood-cleated veneer paper overlaid	Domestic type

Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.1.3.2 Repair parts. - Repair parts shall be packed in containers as specified 5.2.1.3.1.1 and 5.2.1.3.1.2 for the level specified.

5.2.1.4 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.)

6. NOTES

6.1 Intended use. - The pressure gages are intended for measuring boiler casing air pressure.

6.2 Ordering data. - Procurement document should specify the following:

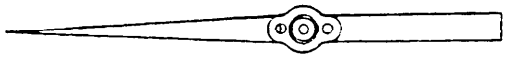
(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Whether connection other than at bottom is required (see 3.6).
(d) Purpose for which gages are intended (see 3.13).
(e) The pressure range of each scale (see 3.14.1.2, 3.15.4, 3.15.5, 3.15.6, 3.15.7, 3.16.2).
(f) Mounting required (see 3.15.6, and 3.16.1).
(g) Size of flared union couplings if other than specified (see 3.16.1).
(h) Number of sets of working and corrected drawings required (see 3.17).
(i) Quantity of repair parts to be furnished (see 3.18.1).
(j) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

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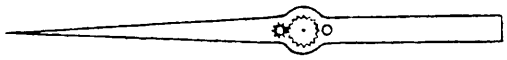
Notice. - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 6685-N030Sh)

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TOP VIEW



SH3385

BOTTOM VIEW

Figure 1 - Micrometer indicating pointer (adjustable).

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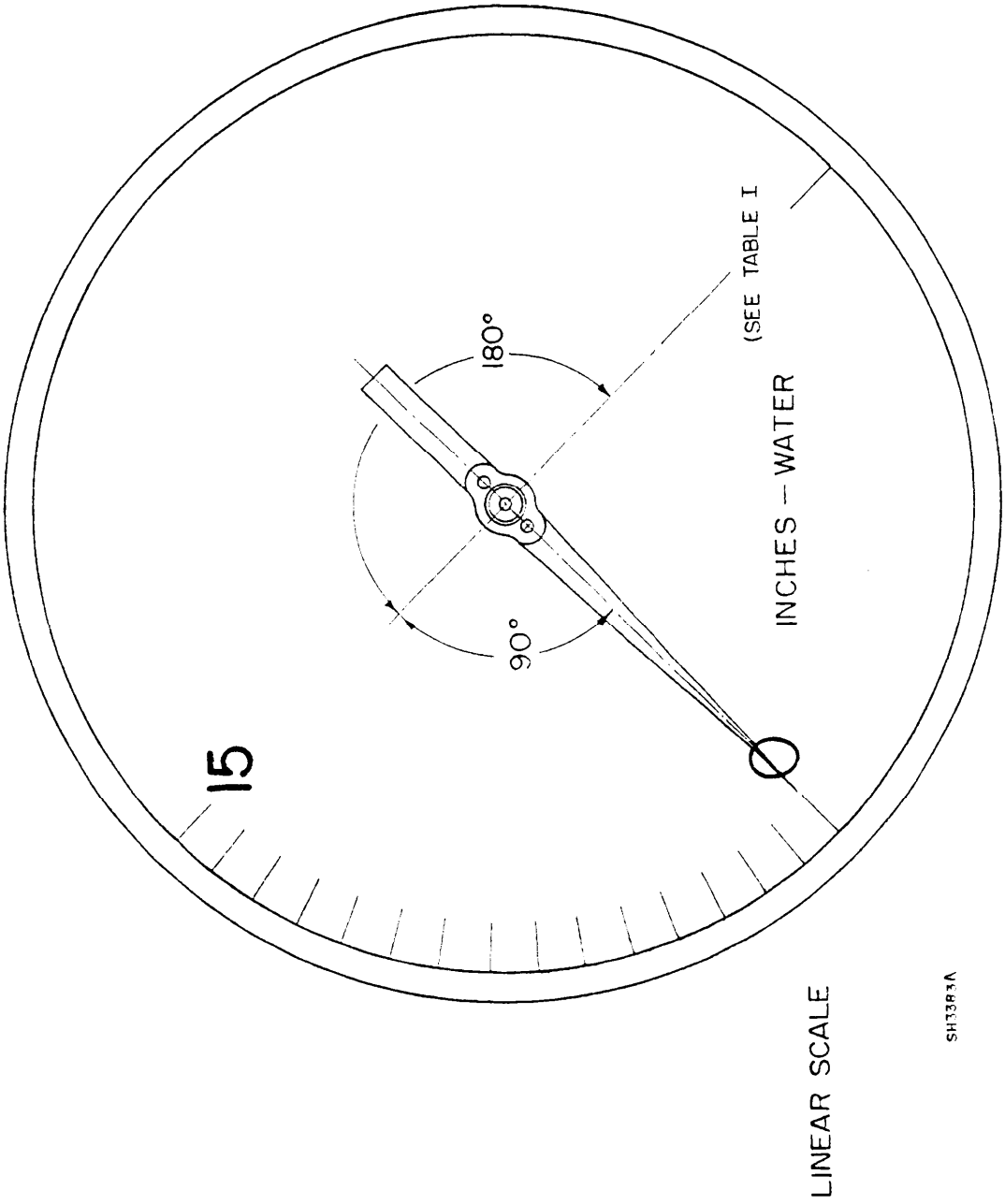


Figure 2 - Dial and indicating pointer.

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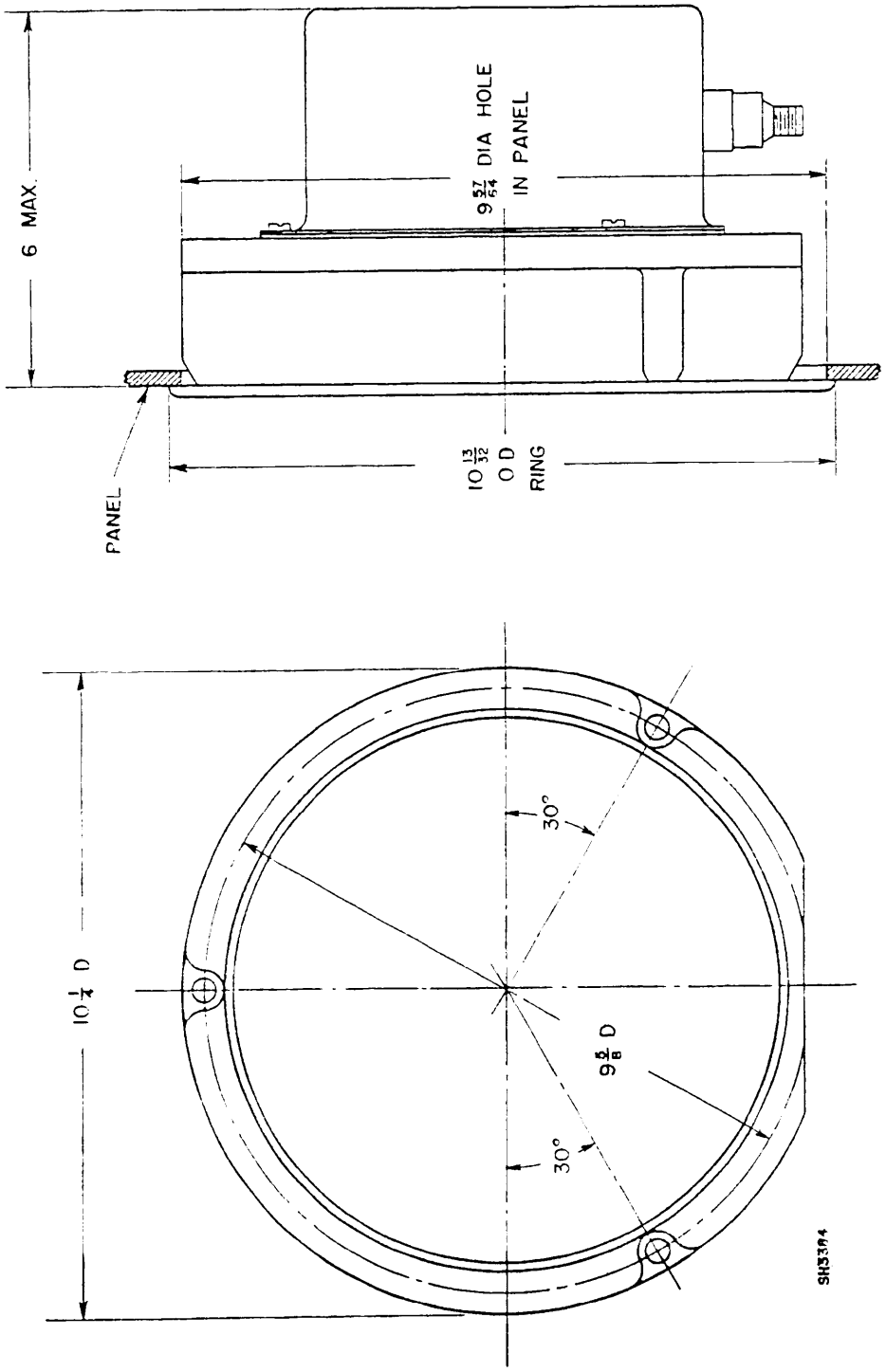
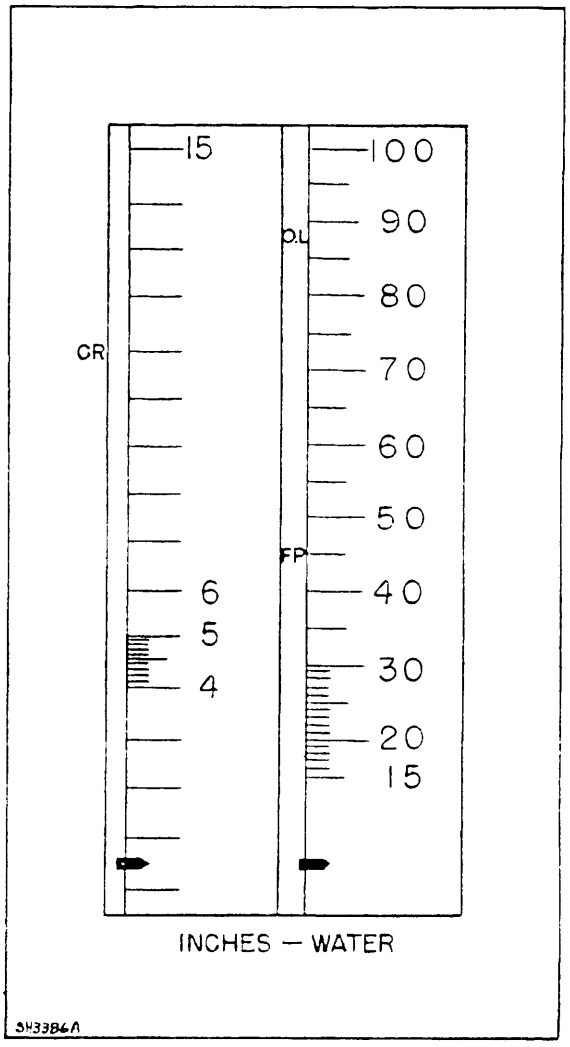


Figure 3 - Type A, 8-1/2 inch gage with extended case.

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LINEAR SCALES

Figure 4 - Scales and indicating pointers.

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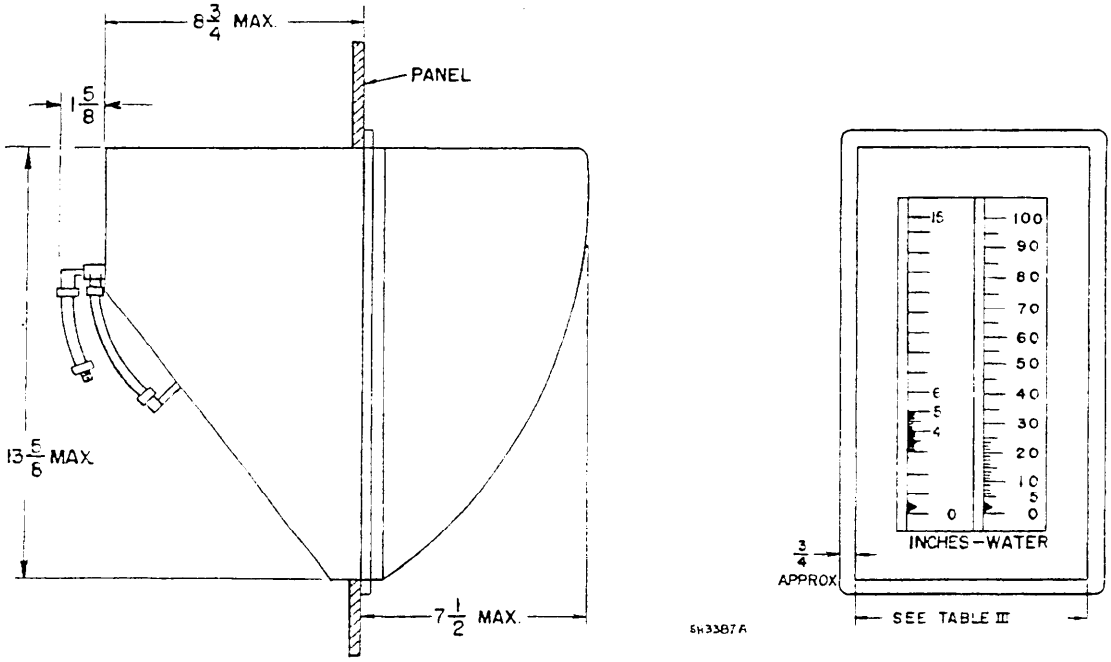


Figure 5 - Type B vertical scale case.

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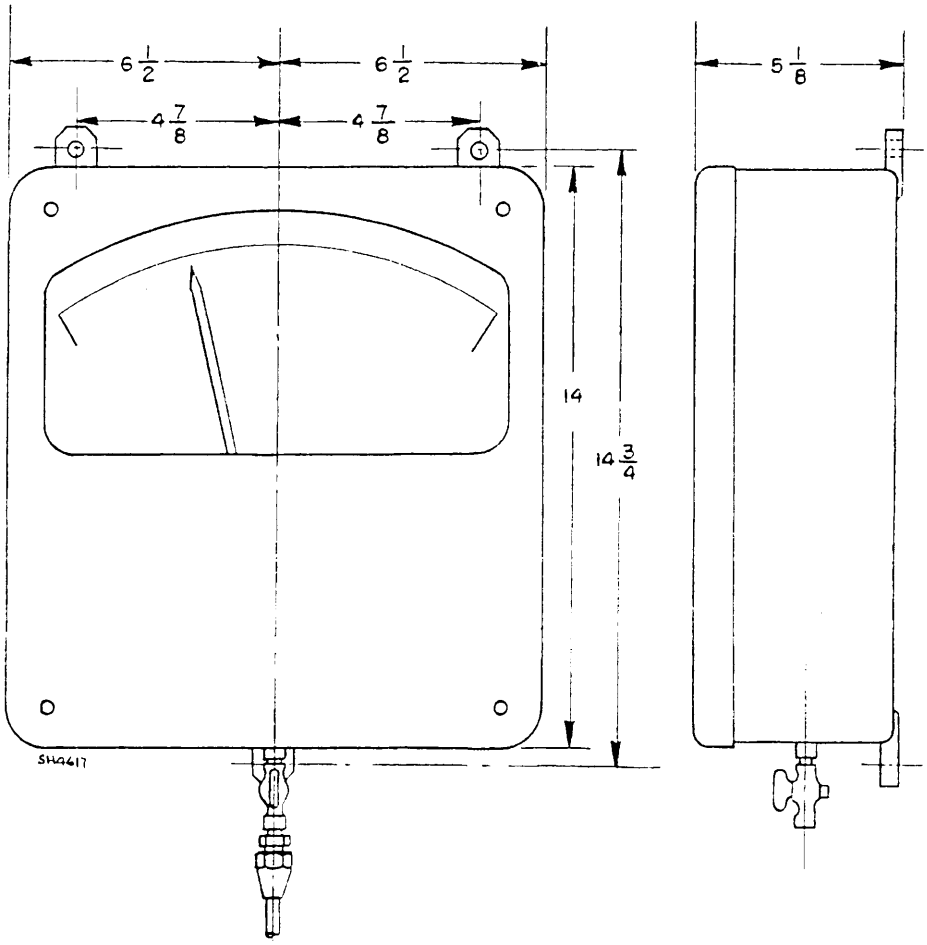


Figure 6 - Type C window gage.

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